Replication and Predictable-Drift Diagnostics for FTS-Diffusion

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Abstract

Financial time-series generators are often evaluated by marginal stylized facts and downstream forecasting gains, but those criteria can miss generator degeneracy, protocol artifacts, and artificial conditional-mean predictability. We revisit FTS-Diffusion, an ICLR 2024 model based on scale-invariant subsequence clustering (SISC), pattern-conditioned diffusion, and learned pattern evolution, and attempt to reproduce it from the released code and configuration. In our hands, the released sampling pipeline produces a near-degenerate generator: synthetic S&P 500 paths are close to a single repeated motif tiled into an almost perfectly linear price ramp (linear-fit $R^2 \approx 1.0$), independent runs are nearly identical, and synthetic daily-return volatility is roughly $20\times$ smaller than real. We trace this empirically to three choices visible in the released sampling code (a deterministic argmax rollout of the pattern chain, a one-unit decoder hidden state, and a commented-out magnitude normalization), while being explicit about confounds we cannot fully rule out: no checkpoints are shipped, and our training budget is reduced relative to the released defaults. Consistent with degeneracy, a continuous-rollout Train-on-Augmentation-Test-on-Real (TATR) setup appears to improve one-day S&P 500 forecasting by up to roughly 85%, yet the same protocol never beats its no-augmentation baseline on GOOG or ZC=F. This non-transfer is what one would expect if the S&P 500 gain were a coincidental scale match between an upward ramp and an upward-trending test period rather than evidence that augmentation helps. We also do not reproduce, on the released artifacts, the authors' return-distribution goodness-of-fit (Kolmogorov–Smirnov and Anderson–Darling tests separate real and synthetic returns in essentially every window), nor the Appendix B.3 SISC toy targets (best per-unit DTW 0.0321 vs. 0.01, best interval IoU 0.6418 vs. 0.784). As a fixed, predeclared extension we add a predictable-drift diagnostic: real S&P 500, GOOG, and ZC=F holdouts lie inside asset-specific block-shuffle null bands; in a 500-seed hidden-drift experiment the statistic tracks exploitable artificial alpha (at sign persistence $p = 0.85$, mean out-of-sample Sharpe 7.72 vs. 0.09 for an iid-sign control); and as a 1000-seed augmentation-selection penalty it lowers held-out normalized MSE from 1.0452 to 1.0087 versus marginal stylized selection. We frame these as empirical observations from one reproduction attempt rather than definitive claims about the method. The cautious lesson is that realistic-looking paths and downstream MAPE on a trending asset are weak evidence on their own: synthetic-finance benchmarks need direct generator diagnostics, protocol-explicit generation, and checks for artificial predictability.

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1 Introduction

Deep generative models are attractive for finance because historical market data are scarce, noisy, nonstationary, and not experimentally reproducible. Financial returns also exhibit heavy tails, weak raw-return autocorrelation, volatility clustering, and recurring shapes at irregular time scales [Cont, 2001]. Sequence generators try to preserve temporal structure with adversarial, optimal-transport, and score-based objectives [Yoon et al., 2019, Xu et al., 2020, Tashiro et al., 2021]; financial generators additionally need to respect stylized facts and economically meaningful dependence [Wiese et al., 2020]. FTS-Diffusion [Huang et al., 2024] addresses this by discovering variable-length motifs and generating new trajectories motif by motif.

This paper is a replication and diagnostic-extension study. We do not introduce a replacement generator. Instead, we test whether the published downstream gains are robust to rollout and evaluation details, whether the SISC toy-data targets are reproducible, and whether a simple predictable-drift statistic catches artificial alpha that marginal stylized facts may miss. We report what we observed in one careful reproduction attempt, and we are deliberately cautious about how far it generalizes. Empirically, the architecture is well motivated, but the artifacts produced by the released sampling pipeline, in our hands and under our compute budget, are degenerate: the synthetic series collapses to a near-linear price ramp built from one repeated motif (Section 4). Under that lens the apparent S&P 500 downstream gain looks less like robust augmentation and more like a coincidence that does not transfer across assets, although we cannot fully separate this observation from a known confound, namely that no trained checkpoints are released and our training budget is shorter than the released defaults.

We therefore extend the evaluation with a predictable-drift diagnostic. Under a fixed information vector

$$\omega_t = (1, r_t, r_{t-1}, |r_t|, |r_{t-1}|)^\top, \tag{1}$$

we measure whether the next return r_{t+1} has a nonzero linear conditional mean under a bounded test class. This is not a proof of market efficiency or absence of arbitrage. It is a finite, reproducible check for an important failure mode: a synthetic generator can match tails and volatility clustering while introducing a simple, exploitable conditional drift.

Contributions. We make five contributions. (i) A code-level diagnosis of the released sampling pipeline (Section 4): we document three specific choices in the released code that, in our runs, drive the synthetic series to a near-degenerate linear ramp, and we record the underspecifications between the paper and the released configuration that we had to resolve. (ii) A six-seed S&P 500 downstream study isolating TATR rollout sensitivity, together with the cross-asset observation that the continuous-rollout gain does not transfer to GOOG or corn futures (ZC=F), which we read as evidence that the S&P 500 result is a scale coincidence. (iii) A return-distribution check that, on the released artifacts, does not reproduce the authors' Kolmogorov–Smirnov and Anderson–Darling goodness-of-fit results. (iv) A documented Appendix B.3 SISC toy-data gap. (v) A fixed predictable-drift diagnostic calibrated on real assets, applied to ordered FTS-Diffusion samples, validated by a 500-seed predictive-alpha experiment, and incorporated in a 1000-seed augmentation-selection benchmark. Throughout, we separate what we can claim about the released artifacts from what would require the authors' checkpoints and full training budget to settle.

2 Background: FTS-Diffusion

FTS-Diffusion decomposes financial time-series generation into three modules [Huang et al., 2024]. Scale-invariant subsequence clustering (SISC) segments a univariate series into variable-length motifs using dynamic time warping (DTW) [Sakoe and Chiba, 1978] and DTW barycenter-style centroid updates [Petitjean et al., 2011]. Each segment is summarized by a pattern identity p_m, a duration scale α_m, and a magnitude scale β_m. A pattern-conditioned diffusion model then generates concrete segments from motif conditions using a DDPM-style denoising objective [Ho et al., 2020].

The long-horizon behavior is controlled by the pattern-evolution module, which learns

$$(p_m, \alpha_m, \beta_m) \mapsto (p_{m+1}, \alpha_{m+1}, \beta_{m+1}), \tag{2}$$

with a classification loss for the next pattern and regression losses for the next duration and magnitude scales. This separation is useful, but it also creates a sensitive interface: downstream results depend on how the learned state chain is initialized, reset, and sliced into synthetic training blocks.

3 Replication protocol

Assets and splits. The main datasets are daily Yahoo Finance close-price histories for S&P 500, GOOG, and ZC=F. The S&P 500 setup spans 1980–2020 and yields 10,088 trading days and 2,282 post-SISC segments with $K = 14$. GOOG spans 2004–2020 with 3,776 trading days and 733 post-SISC segments with $K = 11$. ZC=F spans 2000–2020 with 4,764 trading days and 629 post-SISC segments with $K = 11$. We follow a chronological train/test design: motif extraction, fitted generator components, standardizers, and downstream predictors are fit only on their allowed training windows, and final scores are computed on held-out real data.

Downstream benchmark. The main downstream benchmark is TATR: Train on Augmentation, Test on Real. The held-out real slice is divided into an initialization portion and a final real test portion. A one-day-ahead LSTM forecaster is trained on the real initialization portion plus Y synthetic trading years, where one synthetic year has 252 days and Y is swept across a fixed grid. The score is mean absolute percentage error (MAPE) on the final real test portion. We use paper-like LSTM settings: one-day ahead, window size 64, hidden size 32, two layers, MAE training loss, Adam with learning rate 10^{-2}, and 100 epochs for the primary S&P 500 runs.

Synthetic rollout variants. We compare three ways to produce the synthetic augmentation path.

- **Independent fixed blocks:** generate independent 252-day blocks from the same fixed initial state.
- **Continuous chunked:** generate one continuous trajectory with the pattern-evolution chain, then split it into 252-day chunks for downstream training.
- **Continuous with scaler refit:** generate a continuous trajectory and refit the downstream scaler on the augmented training series.

These protocols use the same broad data budget but different Markov-chain connectivity. The distinction matters because the pattern-evolution module is the only component that carries long-horizon state across generated motifs.

Compute. The primary six-seed S&P 500 downstream batch used stored SISC artifacts and PG-M/PEM checkpoints and recorded $10{,}478$ seconds on an Apple M4 laptop with 10 CPU cores and 16 GB memory. The lightweight drift diagnostics used the same machine: the ordered FTS audit recorded 629.82 seconds with a four-thread cap, the 500-seed predictive-alpha script recorded 2.83 seconds, and the 1000-seed selection benchmark recorded 32.69 seconds. These times exclude earlier exploratory runs not used for the reported tables.

4 Degeneracy of the released generation pipeline

We attempted to reproduce FTS-Diffusion end to end from the released code and configuration. Before discussing downstream scores, we report what the generator actually produces, because this conditions every later interpretation. In our runs, the released sampling path yields a near-degenerate generator: the synthetic price series is close to a single repeated motif tiled into an almost perfectly straight line. We summarize, with file references into our working copy of the released code (`fts-diffusion-ref/`), three choices that appear jointly responsible, then show the degeneracy directly on the saved trajectories and on the return distribution. We stress at the outset that these are observations about the released artifacts under our setup, not proofs about the method; Section 4.5 lists the confounds we cannot rule out.

4.1 Three choices in the released sampling code

(1) Deterministic argmax rollout of the pattern chain. The pattern-evolution module (PEM) is rolled out by taking `argmax` for the next pattern and the next length, and the raw regression mean for the next magnitude, with no sampling (`fts-diffusion-ref/models/sampling.py`, lines 30, 33, 34):

```
pred_pattern   = torch.argmax(probabilities_pattern, dim=1).unsqueeze(0)
pred_length    = torch.argmax(probabilities_length,  dim=1).unsqueeze(0) + l_min
pred_magnitude = pred[:, n_patterns+range_length:].float()
```

An autoregressive map $\text{state}_{m+1} = f(\text{state}_m)$ that is deterministic must, iterated from a fixed initial state, converge to a fixed point or a short cycle. The rollout policy (greedy vs. sampling) is not specified in the paper, so this is a defensible reading of the release, but it removes the stochasticity the Markov-chain description suggests.

(2) One-unit decoder hidden state. The Scaling-AE decoder is an LSTM whose hidden dimension is fed from `pgm_params['sae_hidden_dim']`, set to 1 in the released configuration (`model_params.py`; the decoder LSTM is defined in `scaling_autoencoder.py`). With a single hidden unit the decoder has almost no capacity, and we observe it collapsing to a fixed output that ignores both the conditioning pattern and the diffusion latent.

(3) Commented-out magnitude normalization. The sampling step that would rescale the generated shape to unit range before multiplying by the magnitude β is commented out (`sampling.py`, lines 53–55), so the emitted segment has magnitude $\beta \times$ (whatever range the decoder produced) rather than β. In our runs that range is on the order of 0.07, far below the intended scale, which removes the magnitude information β was meant to carry.

4.2 What the saved trajectories look like

Figure 1 contrasts a real S&P 500 path with one released-pipeline synthetic path. The synthetic series is visually a straight line; a least-squares linear fit gives $R^2 \approx 0.99999$. Three quantitative observations accompany the figure. First, independent runs of a configuration are nearly identical, consistent with a deterministic rollout. Second, the PEM state sequence collapses (Table 1): on S&P 500 a single pattern is used in 99.6% of segments (a fixed point), while GOOG and ZC=F settle into period-2 cycles. Third, inspecting the trained PGM decoder pattern by pattern, the decoder output is near-constant across all patterns (cross-pattern standard deviation < 0.001): an up-trend pattern and a down-trend pattern yield essentially the same segment. The net effect is that one tiny upward-creeping segment is tiled across the horizon, which is exactly the linear ramp in Figure 1.

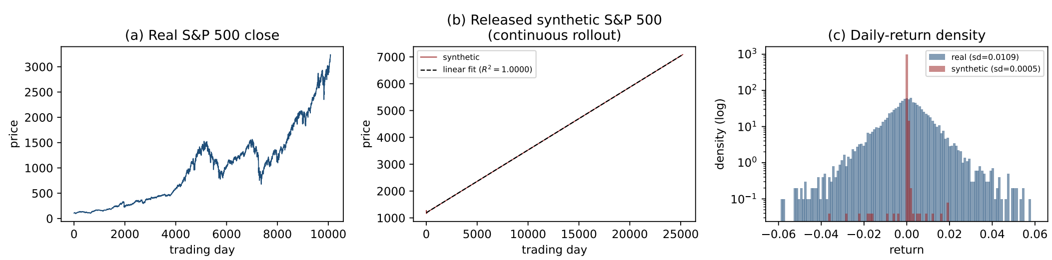

Figure 1: Released-pipeline output on S&P 500. (a) Real close prices show rich structure. (b) One synthetic continuous-rollout path is almost perfectly linear (linear-fit $R^2 \approx 0.99999$). (c) Daily-return density (log scale): synthetic returns concentrate near a single value with standard deviation about $20\times$ smaller than real returns. We show one representative run; runs are nearly identical.

4.3 The return distribution does not match real

A natural and direct check is the authors' own return-distribution test. Following their `distribution_tests` routine, we form simple returns and compare real against synthetic with a Kolmogorov–Smirnov (KS) two-sample test and an Anderson–Darling (AD) k-sample test, where a higher p-value means the two return samples are harder to distinguish. The paper reports values consistent with indistinguishability (KS around 0.32, AD around 0.12). On the released artifacts, in our hands, we observe the opposite (Table 2): across 300 windows per asset and protocol the

Table 1: PEM state-sequence collapse under the deterministic rollout, from saved `run_*_states.npy`. "Dominant pattern share" is the fraction of generated segments using the most frequent pattern.

Asset (K)	Segments	Dominant pattern share	Regime
S&P 500 (14)	2520	99.6%	fixed point
GOOG (11)	2520	50.0%	period-2 cycle
ZC=F (11)	1867	50.0%	period-2 cycle

synthetic and real returns are separated in essentially every window, with mean KS p-values at the 10^{-4} to 10^{-7} level and no window passing at $p > 0.05$. The mechanism is visible in the return moments: the released synthetic series has 4 to $60\times$ too little return volatility, because a near-linear ramp has almost no daily variation. We could not reproduce the published goodness-of-fit numbers from the released artifacts.

Table 2: Return-distribution check on the released artifacts. "Mean KS p" and "% indist." are over 300 windows of length 60; higher is more similar (the paper reports KS around 0.32). "Real/Syn. sd" are daily-return standard deviations. `single` is a continuous rollout, `authors` the independent-block protocol.

Asset	Protocol	Real sd	Syn. sd	Mean KS p	% indist.
S&P 500	authors	0.0109	0.0045	9×10^{-4}	0
S&P 500	single	0.0109	0.00046	4×10^{-7}	0
GOOG	authors	0.0184	0.0026	6×10^{-5}	0
GOOG	single	0.0184	0.00029	4×10^{-7}	0
ZC=F	authors	0.0181	0.0028	2×10^{-5}	0
ZC=F	single	0.0181	0.00041	1×10^{-6}	0

4.4 Underspecifications we had to resolve

Several choices that matter for the result are not pinned down by the paper. Table 3 lists the ones we encountered, separating the three that are load-bearing for the degeneracy above from the compute-driven confounds and the remaining setup choices. We kept the released values wherever they existed, so that the run reflects the artifact as published rather than our preferences. We note this is an audit aid, not an accusation: underspecification is common, but here it coincides with the failure modes.

Table 3: Paper-vs-release underspecifications we resolved. "Released" marks a value shipped in the released configuration that we kept; "ours" marks a choice we had to make.

Design choice	In paper?	Value used	Note
Scaling-AE decoder hidden dim	no	1 (released)	load-bearing
PEM rollout policy	no	argmax (released)	load-bearing
Magnitude normalization at sampling	implied	commented out (released)	load-bearing
PGM / PEM training epochs	no	30 / 60 (ours; released 400 / 1000)	confound
PCDM diffusion steps	no	30 (ours; released 100)	confound
SISC k-means seed	no	42 (released)	shared seed
GOOG / ZC=F downstream split	S&P only	62.5% train (ours)	cross-asset parity
Initial segment for generation	no	first test segment / random (ours)	protocol switch

4.5 Confounds and what we cannot rule out

We are explicit that the evidence above concerns the released artifacts as configured and as trained in our hands, not the method in principle. Two confounds prevent a stronger claim. First, no trained checkpoints are shipped with the release, so every reproduction retrains from the released scripts and we cannot compare against the authors' own generations. Second, our training budget is reduced relative to the released defaults (PGM 30 vs. 400 epochs, PEM 60 vs. 1000, PCDM 30 vs. 100 steps), a compute-driven choice that a reviewer can reasonably challenge. The degeneracy is therefore consistent with at least three non-exclusive explanations that we have not fully separated: undertraining, an unlucky single seed, and the one-unit decoder capacity. A full-budget retrain across seeds would falsify or confirm the undertraining hypothesis; we flag it as the natural next experiment and do not assert a method-level conclusion without it. Given how unusual a one-unit decoder hidden state is for a published generative model, it would also be reasonable to ask the authors directly whether that value, and the commented-out magnitude normalization, correspond to the configuration behind the paper's headline results.

5 Replication results

5.1 TATR depends on the synthetic-series protocol

Our current S&P 500 evidence has three layers. First, the stored result matrices do not support a monotone-downward TATR claim for the documented `authors` independent-block protocol: the stored S&P 500 `authors` curve finishes 208.4% worse than its baseline, while the `single` continuous-trajectory curve finishes 71.6% better. Second, a one-seed protocol search found that continuous-rollout variants finish 76.5–85.4% better, whereas independent fixed blocks finish 96.5% worse. Third, Table 4 reports the current primary six-seed long batch.

In that long batch, continuous protocols reproduce strong paper-like reductions in downstream MAPE, but independent fixed blocks do not. The conclusion is not driven by a single lucky seed: the final continuous-chunked changes range from -86.8% to -55.1%, while the final independent-block changes range from $+37.1\%$ to $+296.0\%$. The contrast is also visible in Figure 2: continuous trajectories eventually cross below the real-only baseline, while independent blocks worsen as augmentation increases.

Table 4: Six-seed S&P 500 TATR summary. Percentages are relative to each protocol's no-augmentation baseline; lower MAPE is better. Ranges are seed-wise final changes.

Protocol	Baseline	Best change	Final change	Final range
Continuous chunked, 1-day	0.0589	-84.9%	-65.2%	$[-86.8, -55.1]\%$
Continuous + scaler refit, 1-day	0.0630	-85.4%	-68.2%	$[-84.3, -30.1]\%$
Independent fixed blocks, 1-day	0.0646	0.0%	$+193.2\%$	$[+37.1, +296.0]\%$
Continuous chunked, 5-day	0.0810	-85.2%	-77.5%	$[-85.9, -58.5]\%$

The mechanism is visible in the synthetic price levels. In the long-batch diagnostics, continuous synthetic trajectories start near 1253 and end near 7085 on average, with mean price level 4138. Independent blocks start from the same level but repeatedly reset, ending near 1249 with mean level 1214. The continuous protocol therefore drifts through the scale of the real test period, while independent blocks remain near the early initial level. Read together with the degeneracy of Section 4, we interpret this as a scale effect rather than evidence of useful augmentation. The continuous protocol drifts a near-linear ramp through the price range of the upward-trending S&P 500 test period, so the forecaster's percentage error can fall simply because synthetic and real price levels overlap, not because the synthetic dynamics are informative. The independent-block protocol keeps the same degenerate dynamics near the initial level, so the overlap does not occur and the error grows. The cross-asset evidence below is the cleaner test of this reading.

TMTR controls make the replication conclusion more conservative. In the six-seed long batch, the reference TMTR curve reaches a best mean improvement of 26.8% but finishes 359.7% worse; a continuous offset-30 control reaches a best improvement of 50.0% and finishes 17.8% better; a continuous offset-50 control finishes 752.0% worse. Thus the evidence is not "synthetic data always helps." It is narrower: a particular continuous TATR price-level rollout can create large apparent gains, while other augmentation and evaluation choices do not support the same conclusion.

We interpret this cautiously and make no claim of misconduct in the original work; the available generation details do not pin down the downstream result uniquely. We note only that, once the generator is seen to be degenerate, the most economical reading of the S&P 500 gain is a scale coincidence rather than augmentation that works under the right protocol. Either way, a reproducible synthetic-finance benchmark should specify the Markov-chain rollout, initialization, block slicing,

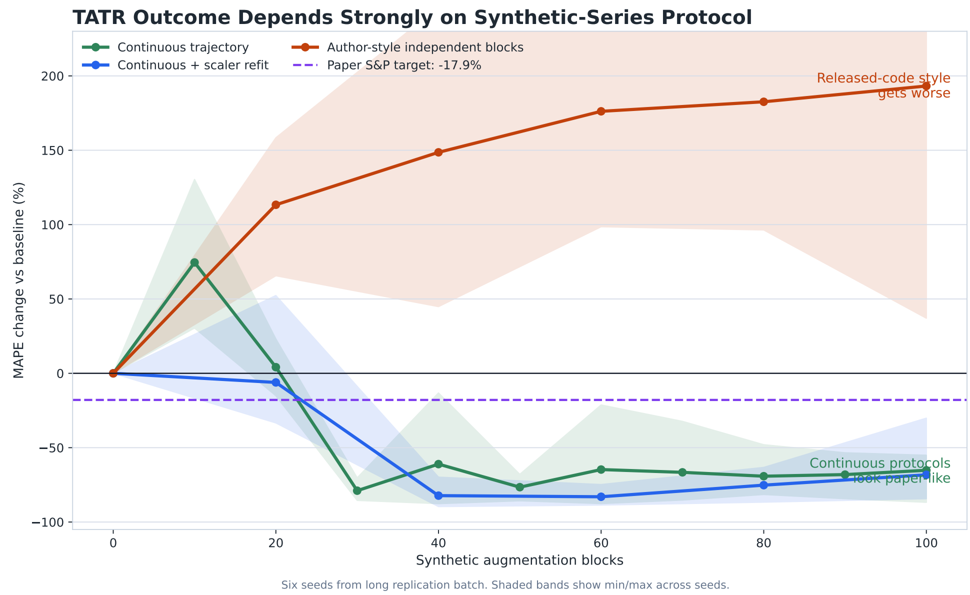

Figure 2: TATR outcome changes sharply with the synthetic-series protocol. Continuous trajectories can look paper-like, while independent blocks worsen with augmentation. The shaded bands show min/max across seeds.

scaler fitting, and warm-up windows as first-class experimental parameters, and should not rest a positive conclusion on downstream MAPE for a single trending asset.

The GOOG and ZC=F replication boundary is also important, and it is where the coincidence reading is tested most directly. We regenerated the stated real-asset SISC and architecture artifacts for GOOG and ZC=F with $K = 11$, $l_{\min} = 10$, and $l_{\max} = 21$. A strict author-faithful downstream TATR rerun is blocked by the released S&P-style split logic, which consumes $252 \times 5 = 1260$ held-out initialization points whereas the GOOG and ZC=F 80/20 held-out windows contain 744 and 963 points, so we adapted the split while keeping the proportions comparable to S&P 500. We emphasize that this adaptation changes the absolute MAPE level but not the sign of the augmentation effect, because every percentage is computed within the same split relative to that protocol's own no-augmentation baseline. Under that within-split comparison the augmentation effect is negative on both assets and for both protocol families. The independent-block (`authors`) protocol finishes 1003.1% worse on GOOG and 75.1% worse on ZC=F. Crucially, the *continuous* rollout that appears to help on S&P 500 does not transfer either: its best point over the whole augmentation sweep equals the no-augmentation baseline (it never beats it), and it finishes about 106% worse on GOOG and 179% to 431% worse on ZC=F depending on the continuous variant. A protocol that helped only because the generator carried useful structure should not flip sign across assets. The fact that the continuous gain is confined to the one strongly trending asset is, in our reading, the signature of a scale coincidence rather than of augmentation value. We still treat GOOG and ZC=F as covered for SISC/architecture setup and real-asset diagnostics rather than as author-faithful downstream replications, but the within-split sign of the effect does not depend on author-faithfulness.

5.2 SISC toy-data replication gap

We also stress-tested SISC on the Appendix B.3 simulated-data setting. The paper target for the multi-pattern toy experiment is average per-unit DTW 0.01 and segmentation Jaccard/IoU 0.784. Across 18 successful runs varying seed, $l_{\max}$, and maximum iterations, our best average per-unit DTW was 0.0321, and our best author-style interval IoU was 0.6418. Figure 3 summarizes the gap.

The gap persisted when increasing SISC iterations from 10 to 20 and when changing $l_{\max}$ from 20 to 21. The best strict boundary Jaccard at tolerance 2 was 0.3608, and the best reference-code-style interval IoU was 0.6418. The best centroid metric remained about $3.2\times$ the reported target. This does not invalidate SISC, but exact B.3 reproduction appears to require more detail about the synthetic

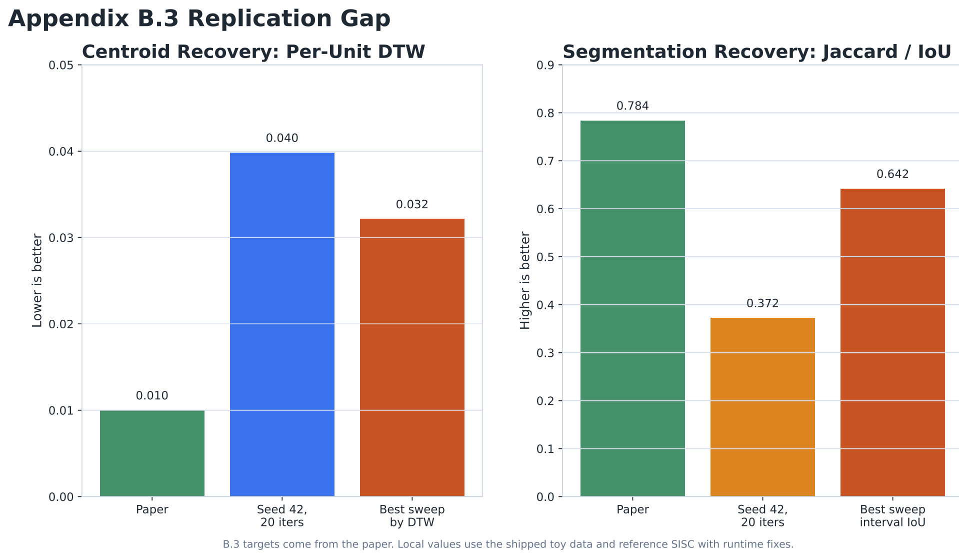

Figure 3: Appendix B.3 SISC replication gap on the toy data. The best replication runs do not reach the reported centroid or segmentation targets.

generator, seed state, one-pattern construction, and metric definitions. The broader replication lesson is that helper defaults matter: warm-up windows, augmentation scales, plotting conventions, and interval-overlap metrics can all change the apparent result if they are not audited.

6 Predictable-drift extension

6.1 Diagnostic

Stylized facts are necessary but not sufficient for finance. A generator can reproduce heavy tails and volatility clustering while encoding a simple conditional mean, such as $r_{t+1} = \rho r_t + \epsilon_{t+1}$. Marginal tests may not detect this, but a trading signal using r_t can. We therefore treat the drift check as a fixed, predeclared diagnostic rather than a learned discriminator: the chronological split rule, feature map, train-only standardization, null construction, and companion stylized-fact summaries are fixed before evaluating synthetic outcomes.

Motivated by the martingale-difference intuition behind weak-form market efficiency [Fama, 1970], we fix a small information set

$$\omega_t = (1,\ r_t,\ r_{t-1},\ |r_t|,\ |r_{t-1}|)^\top \tag{3}$$

and a bounded linear class

$$g_b(\mathcal{F}_t) = b^\top \omega_t, \qquad \|b\|_2 \le 1. \tag{4}$$

The population predictable-drift violation is

$$\delta = \sup_{\|b\|_2 \le 1} \left| \mathbb{E}\left[r_{t+1} b^\top \omega_t \right] \right|. \tag{5}$$

Let $m = \mathbb{E}[r_{t+1}\omega_t]$. By Cauchy-Schwarz, Equation 5 has the closed form

$$\delta = \|m\|_2\,. \tag{6}$$

The empirical estimator is therefore

$$\hat{\delta} = \left\| \frac{1}{T-2} \sum_{t=2}^{T-1} r_{t+1}\omega_t \right\|_2. \tag{7}$$

The statistic decomposes into interpretable components: unconditional drift, first-lag return predictability, second-lag return predictability, and two volatility-conditioned mean terms. In our implementation, returns are standardized using training-split mean and standard deviation only; the constant feature remains equal to one. We compare real data, synthetic data, and a block-shuffled null that preserves the return multiset and some local dependence while disrupting longer chronological alignment.

6.2 Real-asset calibration

Before using $\hat{\delta}$ as a rejection statistic, we calibrate it on real held-out assets. For S&P 500, GOOG, and ZC=F, we start from raw close-price CSVs, use the first 80% of simple returns for standardization, and evaluate the last 20%. The null distribution uses 500 replications of 21-day block shuffling on the held-out returns. Table 5 shows that all three real held-out paths lie inside their asset-specific 5–95% null bands, while retaining heavy tails and volatility clustering. This is a useful sanity check: the diagnostic is not simply rejecting real financial data because it is non-Gaussian.

Table 5: Real-asset predictable-drift calibration. The null band is the 5–95% range from 500 block-shuffled held-out paths; the final columns report stylized-fact context.

Asset	Test obs.	Real $\hat{\delta}$	Null 5–95%	ACF($\lvert r\rvert$)	Ex. kurt.
S&P 500	2018	0.0297	[0.0222, 0.0303]	0.2165	3.2110
GOOG	755	0.0575	[0.0403, 0.0629]	0.0935	6.1142
ZC=F	953	0.0173	[0.0150, 0.0259]	0.1250	2.0247

6.3 Controlled predictive-alpha test

We next test whether larger $\hat{\delta}$ corresponds to exploitable predictability. In each of 500 seeds, we simulate a zero-mean heavy-tailed GARCH-like path, keep the evaluation-period magnitude path $|r_t|$ fixed, and replace only the signs. The iid-sign control has no intentional sign predictability; Markov-sign samples have sign persistence p. We fit a ridge linear predictor on the first half of each evaluation path using the same feature map ω_t, then evaluate one-step prediction and a sign strategy on the second half.

Table 6 and Figure 4 show the result. The iid-sign and real controls have near-zero out-of-sample correlation, near-50% directional accuracy, and near-zero strategy Sharpe. At $p = 0.85$, the hidden-drift sample has mean $\hat{\delta} = 0.5126$, out-of-sample correlation 0.4112, directional accuracy 79.76%, and mean annualized Sharpe 7.72 with seed standard deviation 1.18. Across all controlled samples, pooled $\hat{\delta}$ and out-of-sample Sharpe have correlation 0.6921. Thus the diagnostic is not merely a reporting statistic; it is aligned with artificial alpha.

6.4 Ordered FTS audit and incorporated selection

The FTS-Diffusion use case is direct: report Equation 7 on ordered samples from each rollout protocol, and do not select a protocol only because it improves TATR if its $\hat{\delta}$ lies outside the real/null band. We regenerated six S&P 500 synthetic paths per protocol from stored FTS-Diffusion checkpoints, using 100 ordered 252-day blocks per seed and excluding artificial transitions between independent blocks. Table 7 shows that continuous rollouts stay inside the S&P 500 null band in all six seeds, while independent fixed blocks violate it in all six. The audit therefore does not reject the continuous rollout that produces paper-like TATR gains; it identifies the independent reset protocol as both downstream-harmful and predictably drifted under this statistic.

We next test whether the same principle helps choose augmentations in a controlled benchmark. For each of 1000 seeds, all candidate augmentations shared the same magnitude path but had different sign dynamics. A marginal stylized selector minimized a distance over volatility, tail, quantile, and absolute-return autocorrelation summaries; the drift-incorporated selector minimized

$$d_{\mathrm{styl}} + 0.25\left[\hat{\delta}/q^{\mathrm{real}}_{.95} - 1\right]_+^2, \tag{8}$$

where $q^{\mathrm{real}}_{.95}$ is the real block-shuffle null cutoff. The selected augmentation trained the same one-step ridge predictor on a 252-day real initialization plus synthetic returns, and performance was measured on a final held-out real path that the drift-penalized selector does not see.

Table 6: Controlled predictive-alpha test over 500 seeds. Magnitude paths are fixed within seed; only sign dependence changes.

Sample	p	$\hat{\delta}$	OOS corr.	Dir. acc.	Sharpe (sd)
Real evaluation path	–	0.0976	0.0004	0.5025	0.07 (0.77)
IID signs	–	0.0941	0.0036	0.5020	0.09 (0.74)
Markov signs	0.65	0.2088	0.1482	0.5968	2.33 (0.96)
Markov signs	0.75	0.3435	0.2786	0.6952	4.80 (1.05)
Markov signs	0.85	0.5126	0.4112	0.7976	7.72 (1.18)
Markov signs	0.90	0.6164	0.4811	0.8483	9.43 (1.33)

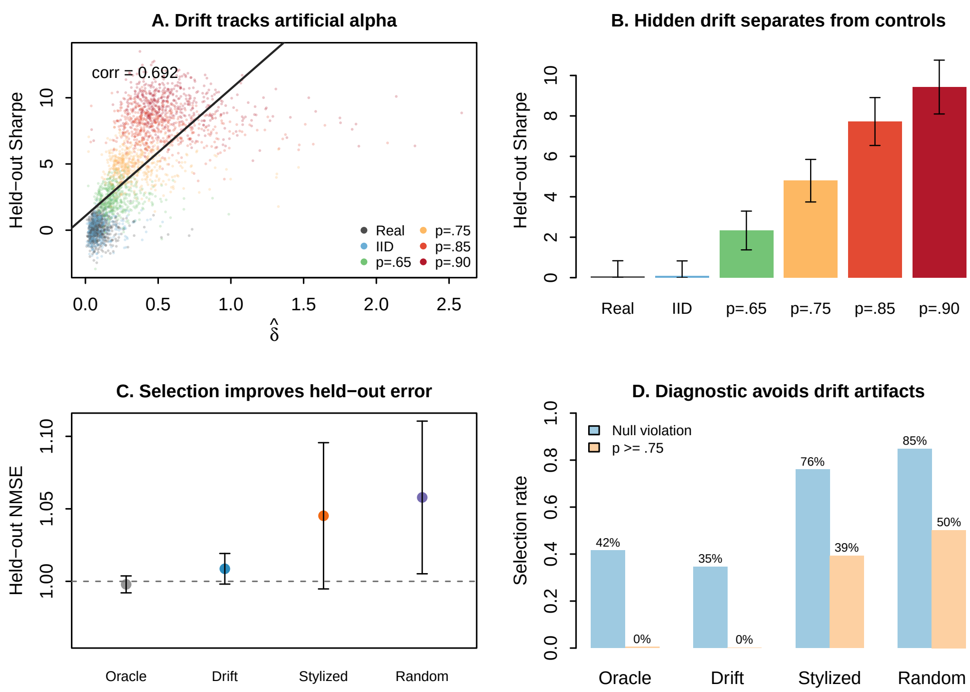

Figure 4: Drift versus no-drift evidence from the larger controlled experiments. The diagnostic tracks held-out artificial alpha, separates hidden-drift sign processes from real and iid controls, and improves augmentation selection by avoiding null violations and high-persistence candidates.

Table 8 shows that incorporating the diagnostic changes the selected augmentation, not just the reporting. Relative to stylized-only selection, the drift-penalized selector reduces high-persistence choices from 39.3% to 0.2%, cuts real-null violations from 76.0% to 34.6%, and lowers held-out normalized MSE by 0.0365 on average, with a 60.8% paired win rate. If the statistic is incorporated during training rather than selection, use it as a validation-selected penalty

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FTS}} + \lambda \hat{\delta}^2_{\text{synthetic}}. \tag{9}$$

The penalty must be computed on ordered return sequences. Applying it across randomly shuffled mini-batches would destroy the time ordering and should only be logged as a proxy, not as the final diagnostic.

7 Limitations and broader impact

Our replication is constrained by the information and compute available for this study. Most importantly, the degeneracy analysis of Section 4 concerns the released artifacts as configured and

Table 7: Ordered S&P 500 FTS-Diffusion drift audit over six seeds and 25,200 generated prices per seed. The cutoff is the S&P 500 held-out block-shuffle null 95th percentile, $q_{.95} = 0.0303$.

Protocol	$\hat{\delta}$	$\hat{\delta}/q_{.95}$	Violations	Price std.
Continuous rollout	0.0263	0.87	0/6	1701.5
Independent fixed blocks	0.0370	1.22	6/6	24.5

Table 8: Drift-incorporated augmentation selection over 1000 seeds. Lower normalized MSE is better. High-p is the fraction of selected Markov-sign candidates with persistence $p \geq 0.75$. The oracle uses held-out test MSE and is only a reference upper bound.

Selector	Null viol.	High-p	Styl. score	Test NMSE	Excess
Oracle test-MSE	41.5%	0.5%	0.2135	0.9979	-0.0319
Drift-penalized	34.6%	0.2%	0.0520	1.0087	-0.0211
Stylized-only	76.0%	39.3%	0.0344	1.0452	$+0.0154$
Random	84.8%	50.2%	0.3106	1.0579	$+0.0281$

as trained in our hands: no checkpoints are shipped, and our training budget is reduced relative to the released defaults. We have not run the full-budget, multi-seed retrain that would separate undertraining from an architectural cause, so we deliberately phrase those findings as observations about the release rather than conclusions about the method. Some runs use reference checkpoints rather than a full fresh retraining of all modules, and the ordered FTS-Diffusion drift audit is limited to the S&P 500 checkpoints and the fixed diagnostic in Equation 7. Train-time validation selection of λ remains to be run before claiming an optimized drift-regularized generator. GOOG and ZC=F have shorter histories than S&P 500, so the downstream split used for S&P 500 is not directly portable without changing the question being asked; we mitigate this by comparing within split, relative to each protocol's own baseline.

The predictable-drift diagnostic is deliberately narrow. It tests a fixed finite-dimensional feature map, not every admissible trading strategy, and it is not a proof of no-arbitrage. Its strength is reproducibility: the statistic is simple, closed-form, and hard to tune after seeing the result.

Synthetic financial data can be beneficial for stress testing, augmentation, and reproducibility, but it can also create misleading evidence if artificial predictability is mistaken for real market structure. Any release of synthetic paths should clearly label them as generated data, document the training period and generation protocol, and avoid implying that downstream forecasting gains translate into tradable performance.

8 Conclusion

FTS-Diffusion has an appealing inductive bias for financial time series: variable-length motifs, pattern-conditioned diffusion, and learned motif evolution. Our reproduction attempt, however, did not yield a working generator from the released artifacts. In our hands the released sampling pipeline collapses to a near-linear ramp built from one repeated motif, the authors' own return-distribution test does not reproduce on those artifacts, and the one downstream setting that appears to improve forecasting does so only on a single strongly trending asset and fails to transfer to GOOG or ZC=F. The most economical reading of that pattern is a scale coincidence rather than useful augmentation. We are careful not to overstate this: we cannot yet separate the degeneracy from undertraining without the authors' checkpoints and full training budget, so we present empirical observations about the release, not a verdict on the method. What the evidence does support is a methodological point. Realistic-looking paths and a favorable downstream MAPE on a trending series are weak evidence on their own. Future synthetic-finance benchmarks should report exact rollout protocols, direct generator diagnostics such as the return-distribution and predictable-drift checks used here, and cross-asset transfer, rather than resting positive conclusions on a single asset.

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